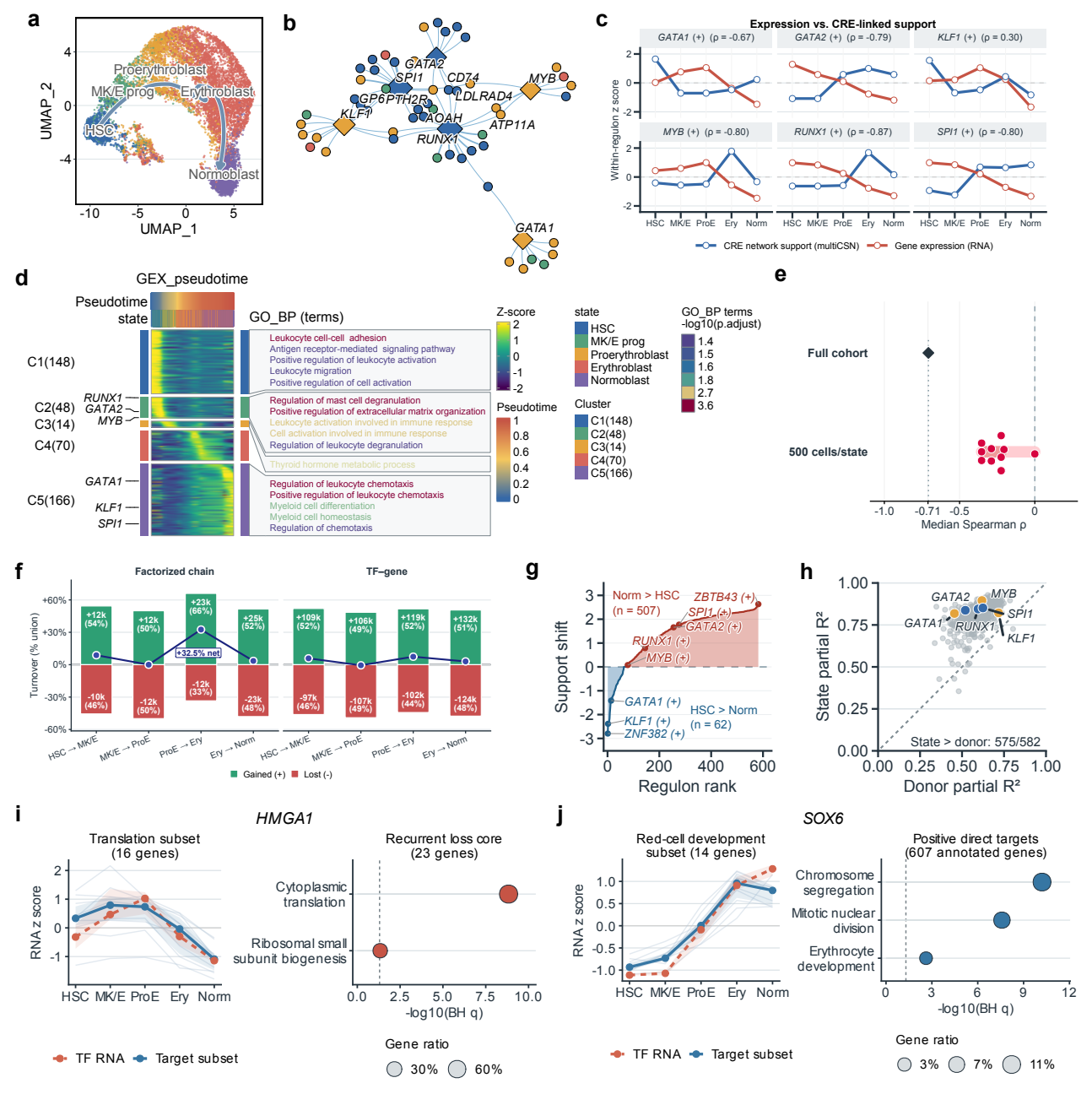

a
Proerythroblast
MK/E prog
Erythroblast
HSC
Normoblast
UMAP_2
UMAP_1
b
GATA2
SPI1
CD74
MYB
GP6
PTH2R
LDLRAD4
KLF1
AOAH
ATP11A
RUNX1
GATA1
c
Expression vs. CRE-linked support
GATA1 (+) (ρ = -0.67)
GATA2 (+) (ρ = -0.79)
KLF1 (+) (ρ = 0.30)
MYB (+) (ρ = -0.80)
RUNX1 (+) (ρ = -0.87)
SPI1 (+) (ρ = -0.80)
Within-regulon z score
HSC MK/E ProE Ery Norm
CRE network support (multiCSN)
Gene expression (RNA)
d
GEX_pseudotime
Pseudotime
state
GO_BP (terms)
C1(148)
C2(48)
C3(14)
C4(70)
C5(166)
RUNX1
GATA2
MYB
GATA1
KLF1
SPI1
Leukocyte cell-cell adhesion
Antigen receptor-mediated signaling pathway
Positive regulation of leukocyte activation
Leukocyte migration
Positive regulation of cell activation
Regulation of mast cell degranulation
Positive regulation of extracellular matrix organization
Leukocyte activation involved in immune response
Cell activation involved in immune response
Regulation of leukocyte degranulation
Thyroid hormone metabolic process
Regulation of leukocyte chemotaxis
Positive regulation of leukocyte chemotaxis
Myeloid cell differentiation
Myeloid cell homeostasis
Regulation of chemotaxis
Z-score
Pseudotime
state
HSC
MK/E prog
Proerythroblast
Erythroblast
Normoblast
Cluster
C1(148)
C2(48)
C3(14)
C4(70)
C5(166)
GO_BP terms
-log10(p.adjust)
e
Full cohort
500 cells/state
Median Spearman ρ
f
Factorized chain
TF–gene
Turnover (% union)
HSC → MK/E
MK/E → ProE
ProE → Ery
Ery → Norm
+32.5% net
Gained (+)
Lost (-)
g
Norm > HSC
(n = 507)
ZBTB43 (+)
SPI1 (+)
GATA2 (+)
RUNX1 (+)
MYB (+)
GATA1 (+)
KLF1 (+)
ZNF382 (+)
HSC > Norm
(n = 62)
Support shift
Regulon rank
h
GATA2
MYB
SPI1
GATA1
RUNX1
KLF1
State partial R²
Donor partial R²
State > donor: 575/582
i
HMGA1
Translation subset
(16 genes)
Recurrent loss core
(23 genes)
RNA z score
Cytoplasmic translation
Ribosomal small subunit biogenesis
-log10(BH q)
TF RNA
Target subset
Gene ratio
30%
60%
j
SOX6
Red-cell development subset (14 genes)
Positive direct targets (607 annotated genes)
Chromosome segregation
Mitotic nuclear division
Erythrocyte development
-log10(BH q)
TF RNA
Target subset
Gene ratio
3%
7%
11%